Linkmodel: gevit and cgevit

This is work in progress.

Parametrization

Hyperparameters

Specification

Hyperparameter spesification and default values

doc `GEVIT link`

hyper

theta1

hyperid `49033`
name `gev tail`
short.name `tail`
initial `0.1`
fixed `FALSE`
prior `pc.egptail`
param `5 -0.5 0.5`
to.theta `function(x, interval = c(REPLACE.ME.low, REPLACE.ME.high)) log(-(interval[1]`
from.theta `function(x, interval = c(REPLACE.ME.low, REPLACE.ME.high)) interval[1] + (`

theta2

hyperid `49034`
name `gev p0`
short.name `p0`
initial `0`
fixed `FALSE`
prior `normal`
param `0 1`
to.theta `function(x) log(x / (1 - x))`
from.theta `function(x) 1 / (1 + exp(-x))`

pdf `gevit`

doc `Complement GEVIT link`

hyper

theta1

hyperid `49035`
name `gev tail`
short.name `tail`
initial `-3`
fixed `FALSE`
prior `pc.gevtail`
param `7 0 0.5`

to.theta function(x, interval = c(REPLACE.ME.low, REPLACE.ME.high)) log(-(interval[1]

from.theta function(x, interval = c(REPLACE.ME.low, REPLACE.ME.high)) interval[1] + (

theta2

hyperid 49036

name gev p0

short.name p0

initial 0

fixed FALSE

prior normal

param 0 1

to.theta function(x) log(x / (1 - x))

from.theta function(x) 1 / (1 + exp(-x))

pdf gevit

Example

```
n <- 3000
x <- rnorm(n, sd = .5)
intercept <- runif(1)
beta.x <- runif(1, 0.5, 1.5)
eta <- intercept + beta.x * x


xi <- -0.3
p.intercept <- inla.link.invgevit(intercept, tail = xi)
prob <- inla.link.invgevit(eta, tail = xi)
size <- 2
y <- rbinom(n, size = size, prob = prob)

r <- inla(y ~ 1 + x,
          data = data.frame(y, x),
          family = "binomial",
          Ntrials = size,
          control.inla = list(cmin = 0, int.strategy = "eb"),
          control.fixed = list(remove.names = "(Intercept)"),
          control.family = list(
              control.link =
                  list(model = "gevit",
                       hyper = list(tail = list(prior = "pcegptail",
                                                param = c(7, -0.5, 0.5)),
                                    intercept = list(initial = 0, param = c(0, 1))))),
          verbose = !TRUE)

summary(r)

round(dig = 3,
      cbind(true = c(p.intercept = p.intercept,  beta.x =  beta.x,  xi = xi),
            estimate = c(p.intercept = r$summary.hyperpar[2,"mean"],
                         beta.x = r$summary.fixed["x", "mean"],
```

```
                        xi = r$summary.hyperpar[1, "mean"])))


## this shows that the intercept is not part of the linear predictor, then also, not the fitte
## values
plot(eta, r$summary.linear.predictor$mean +
          inla.link.gevit(r$summary.hyperpar[2,"mean"],
                          r$summary.hyperpar[1,"mean"]),
     lwd = 3, col = "red", type = "l")
abline(a = 0, b = 1, lwd = 1, col = "blue")


############## same check for 'cgevit' link
##p.intercept <- 1 - inla.link.invgevit(intercept, tail = xi)
##prob <- 1 - inla.link.invgevit(eta, tail = xi)
p.intercept <- inla.link.invcgevit(intercept, tail = xi)
prob <- inla.link.invcgevit(eta, tail = xi)
## to get the same data
y <- size - y

rc <- inla(y ~ -1 + x,
           data = data.frame(y, x),
           family = "binomial",
           Ntrials = size,
           control.inla = list(cmin = 0, int.strategy = "eb"),
           control.family = list(
               control.link =
                   list(model = "cgevit",
                        hyper = list(tail = list(prior = "pcegptail",
                                                 param = c(7, -0.5, 0.5)),
                                     intercept = list(initial = 0, param = c(0, 1))))))
print(round(dig = 3,
            cbind(true = c(p.intercept = p.intercept,  beta.x =  beta.x,  xi = xi),
                  estimate = c(p.intercept = rc$summary.hyperpar[2,"mean"],
                               beta.x = rc$summary.fixed["x", "mean"],
                               xi = rc$summary.hyperpar[1, "mean"]))))
```